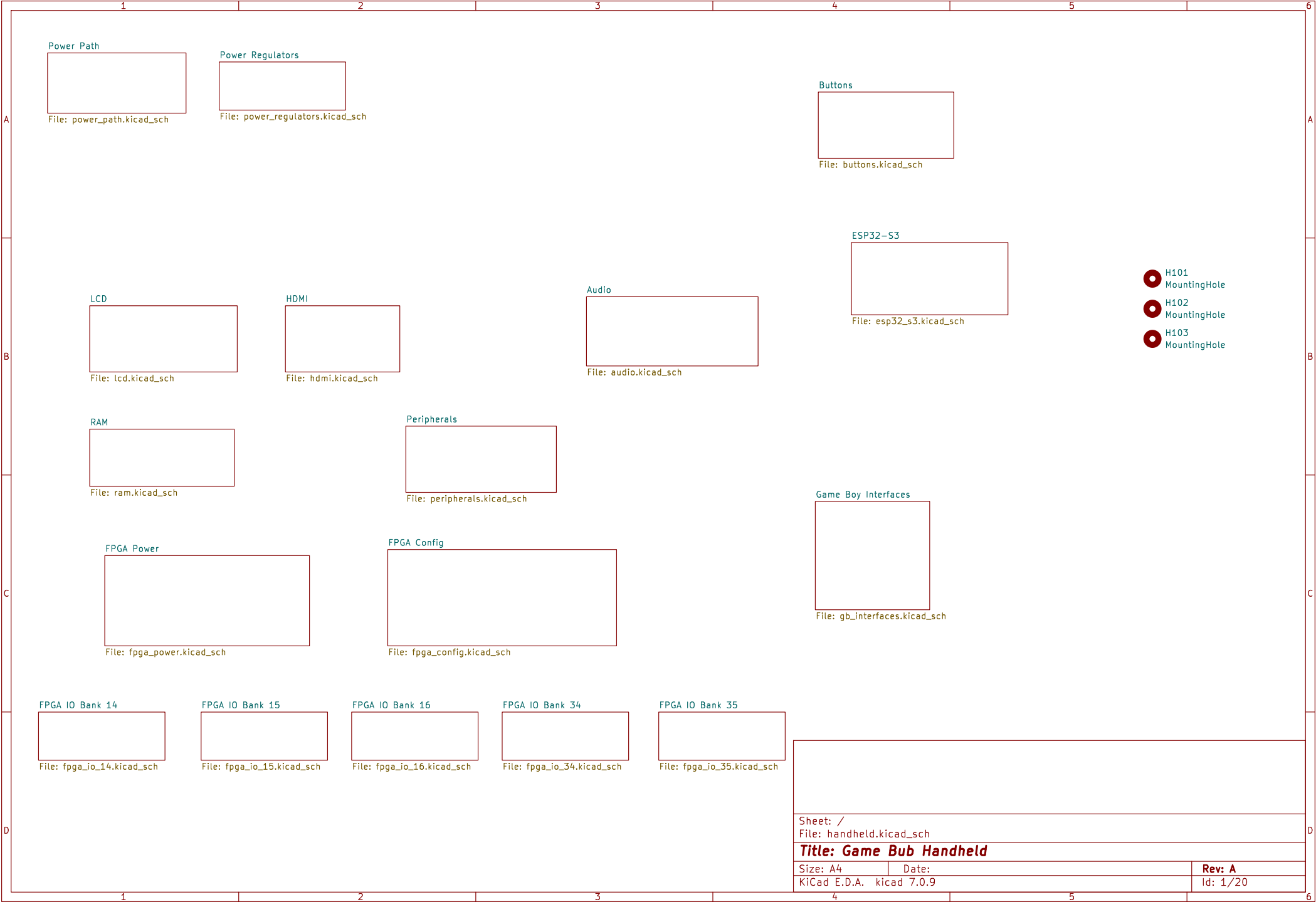
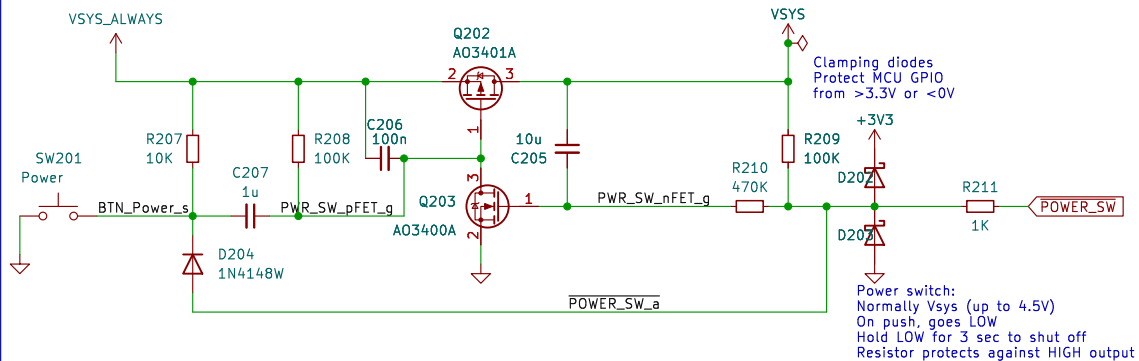
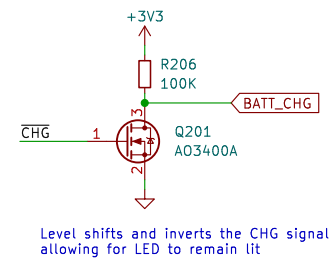
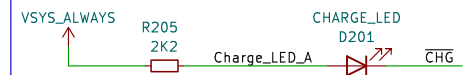
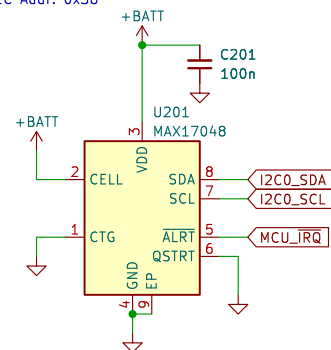
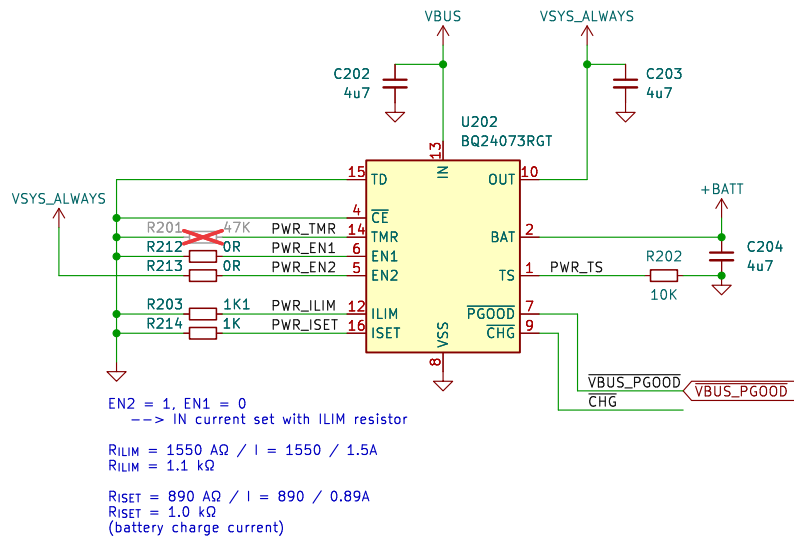
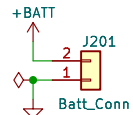
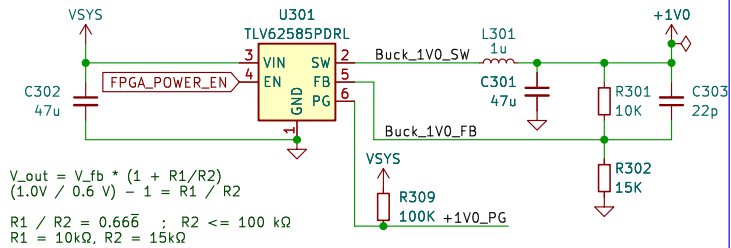




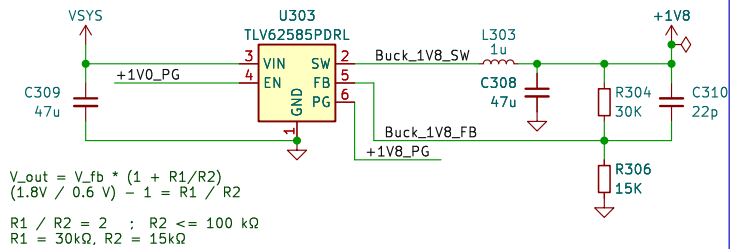


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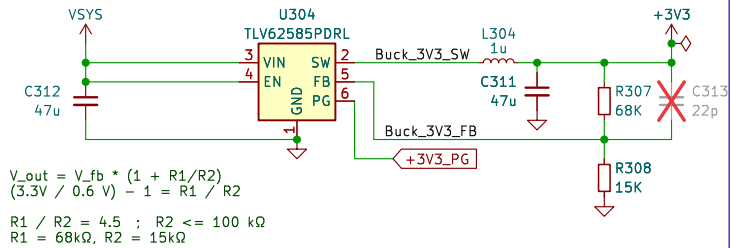
+1V0 Buck



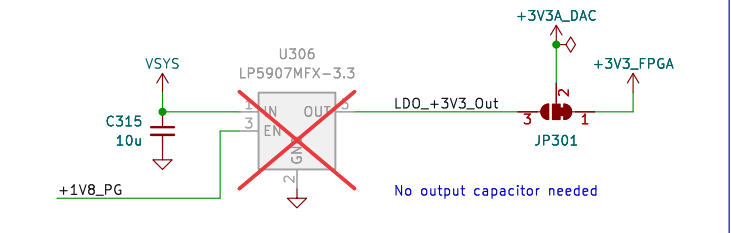
+1V8 Buck



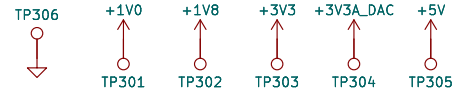
+3V3 Buck



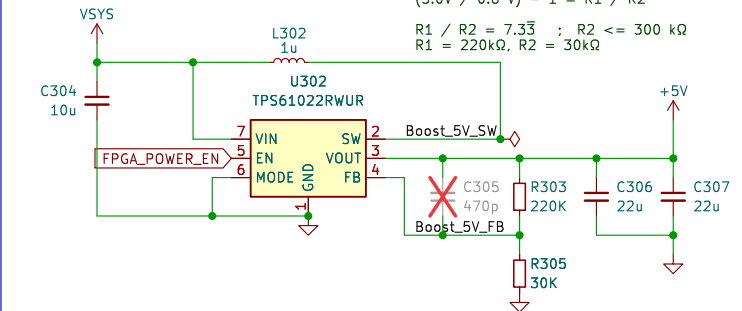
+3V3 Analog DAC



Power Rail Test Points



+5V Boost



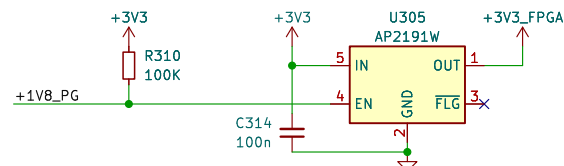
Power Sequencing

Vsys -> +3V3 Enable

FPGA enable:

- > +1V0_FPGA & +5V0
- > +1V8 (FPGA and DAC)
- > +3V3 (FPGA and DAC)

PG signals are open-drain, active high



Sheet: /Power Regulators/
File: power_regulators.kicad_sch

Title:

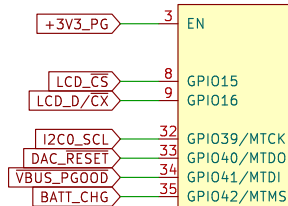
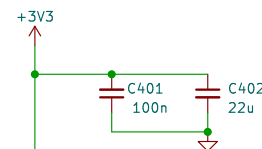
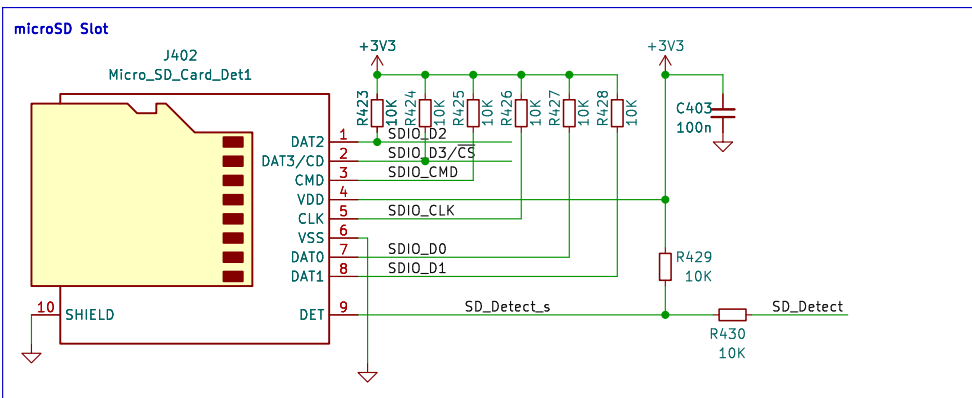
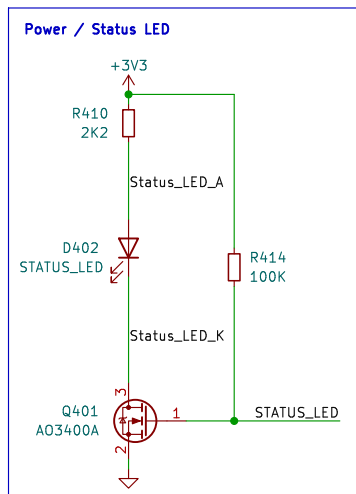
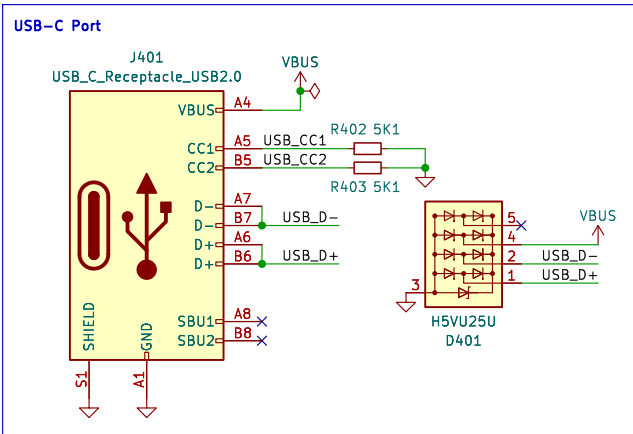
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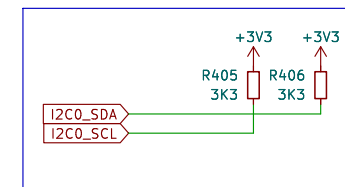
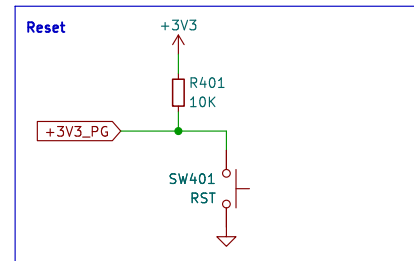
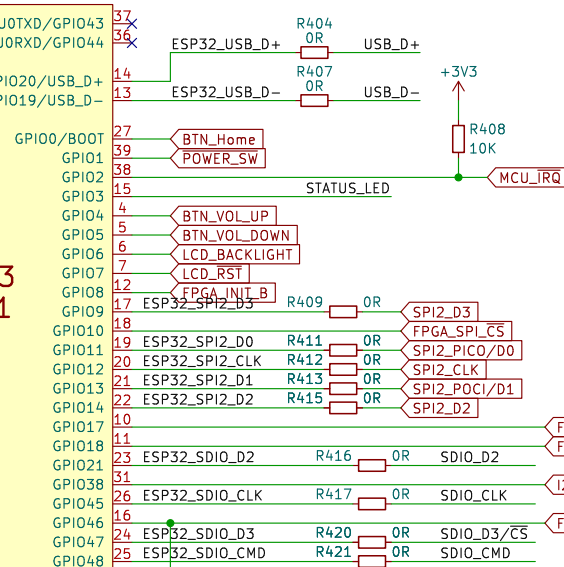
KiCad E.D.A. kicad 7.0.9

Rev:

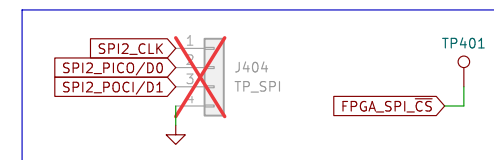
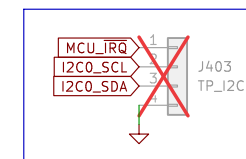
Id: 3/20



**ESP32-S3
WROOM-1**



ESP32-S3 IRQ is open-drain



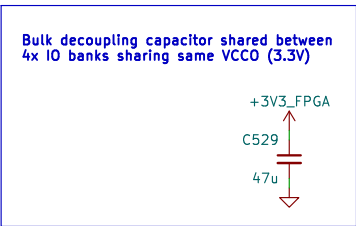
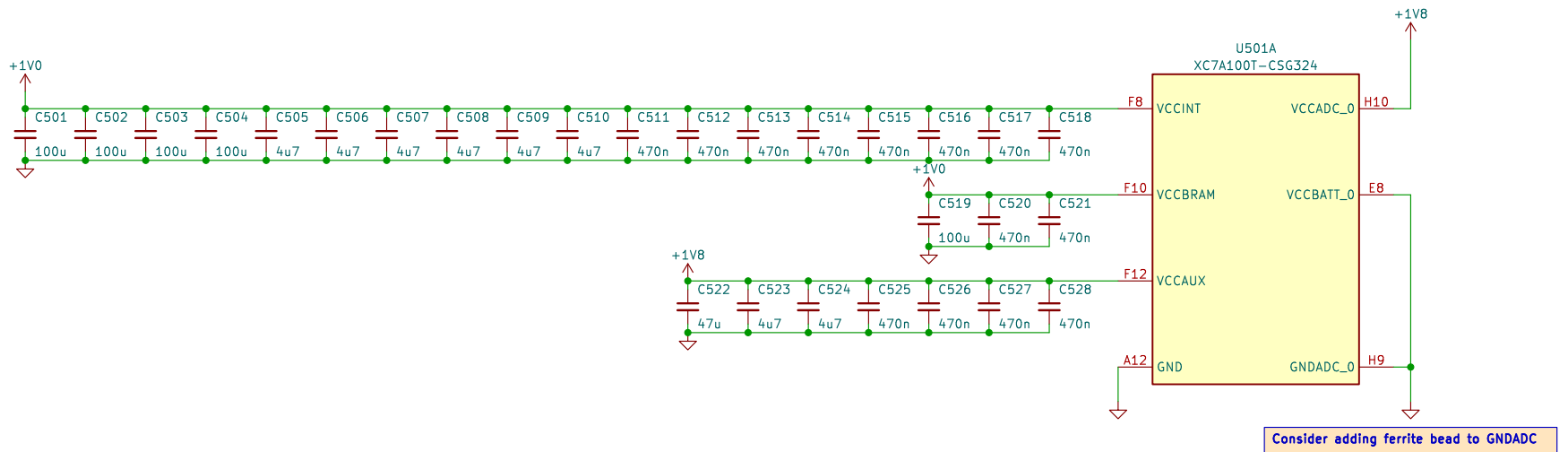
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File: esp32_s3.kicad_sch

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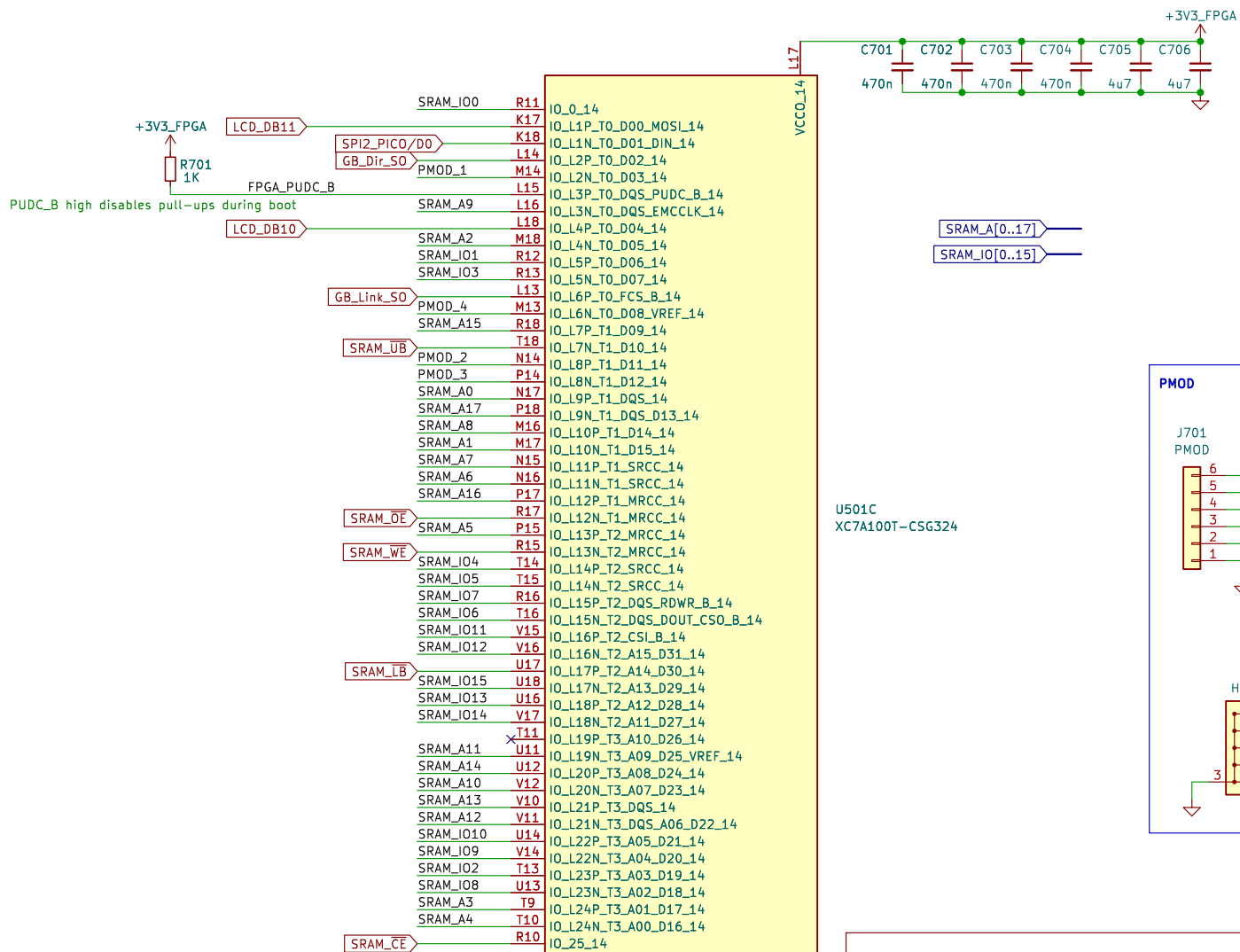
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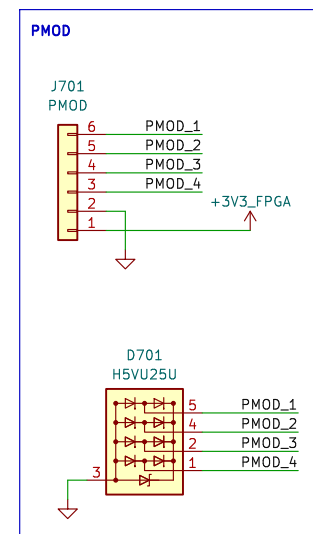
Rev:
Id: 4/20



Sheet: /FPGA Power/ File: fpga_power.kicad_sch		
Title:		
Size: A4	Date:	Rev:
KiCad E.D.A. kicad 7.0.9	Id: 5/20	



U501C
XC7A100T-CSG324



Sheet: /FPGA IO Bank 14/
File: fpga_io_14.kicad_sch

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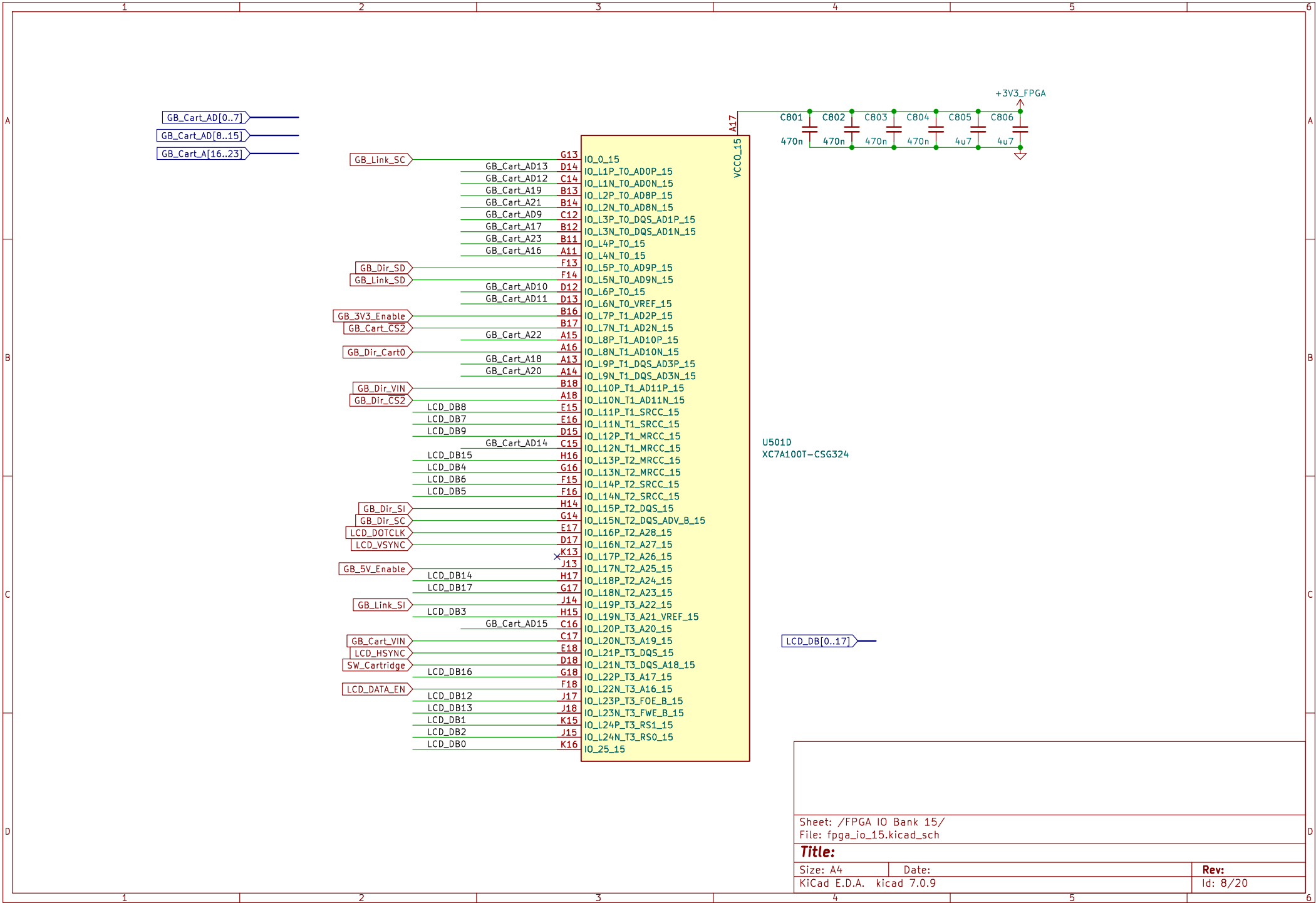
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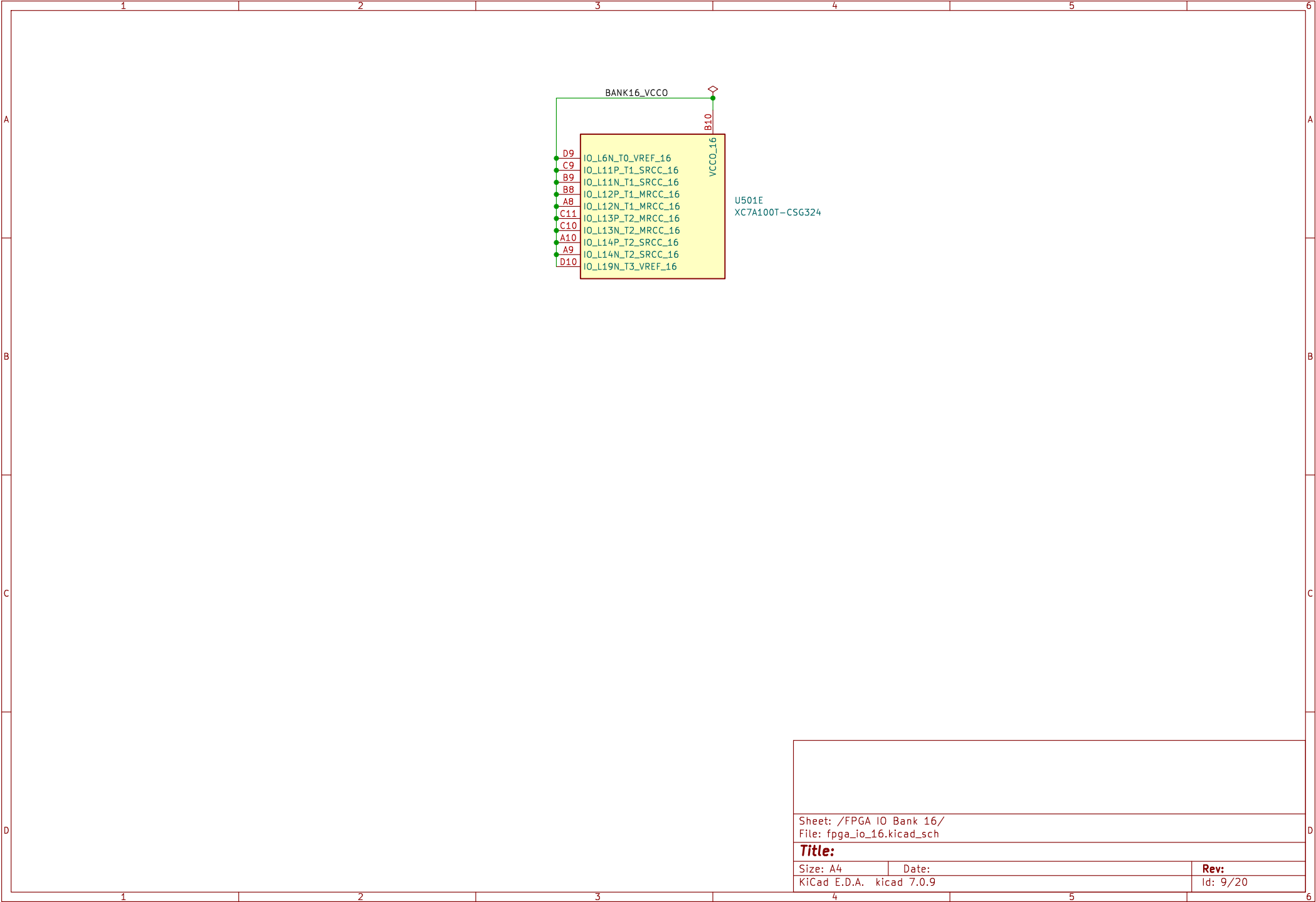
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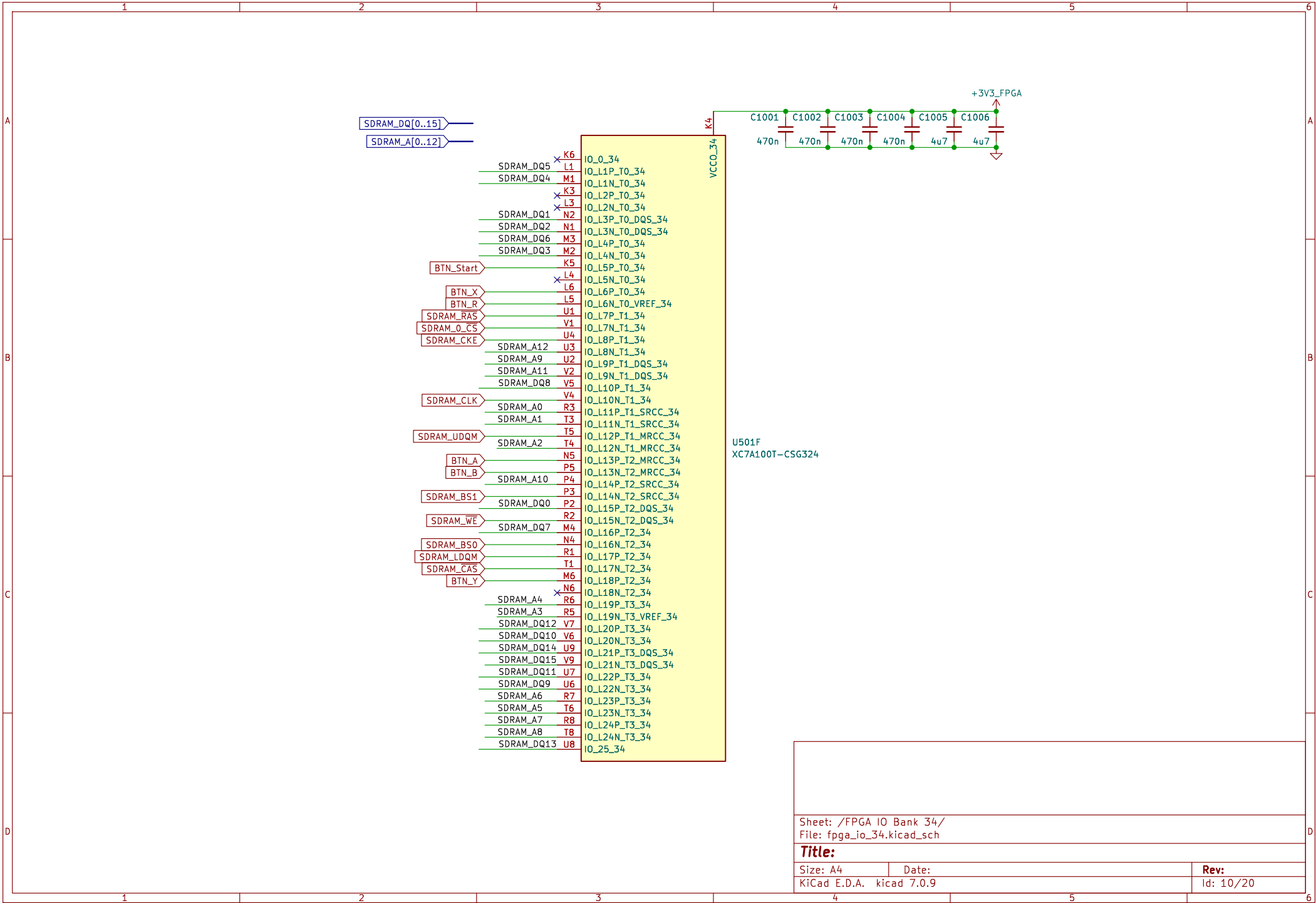
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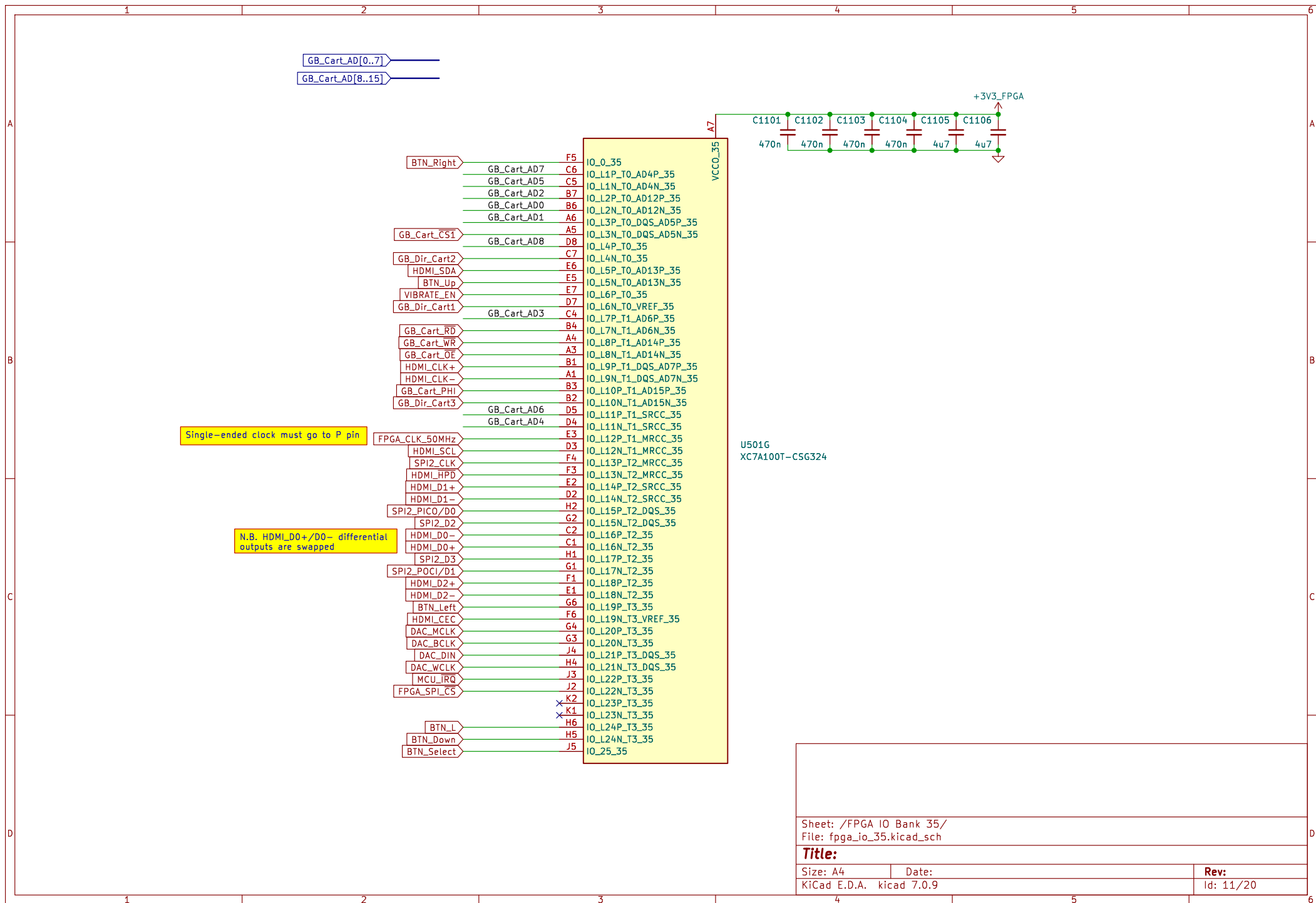
Rev:

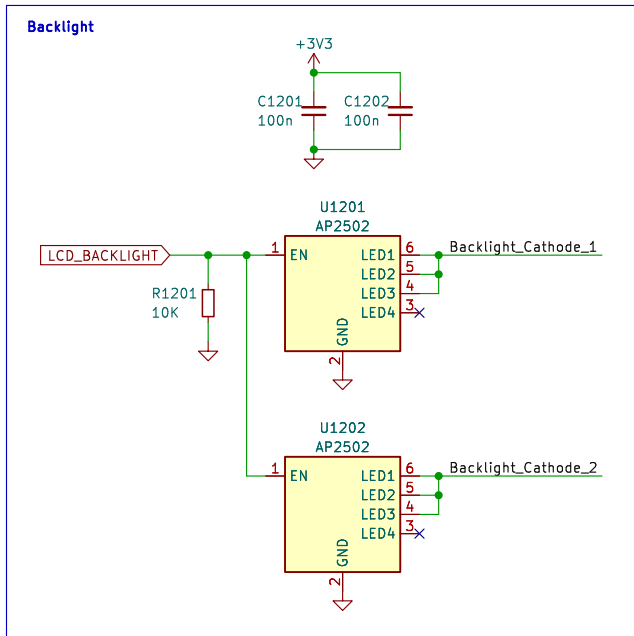
Id: 7/20





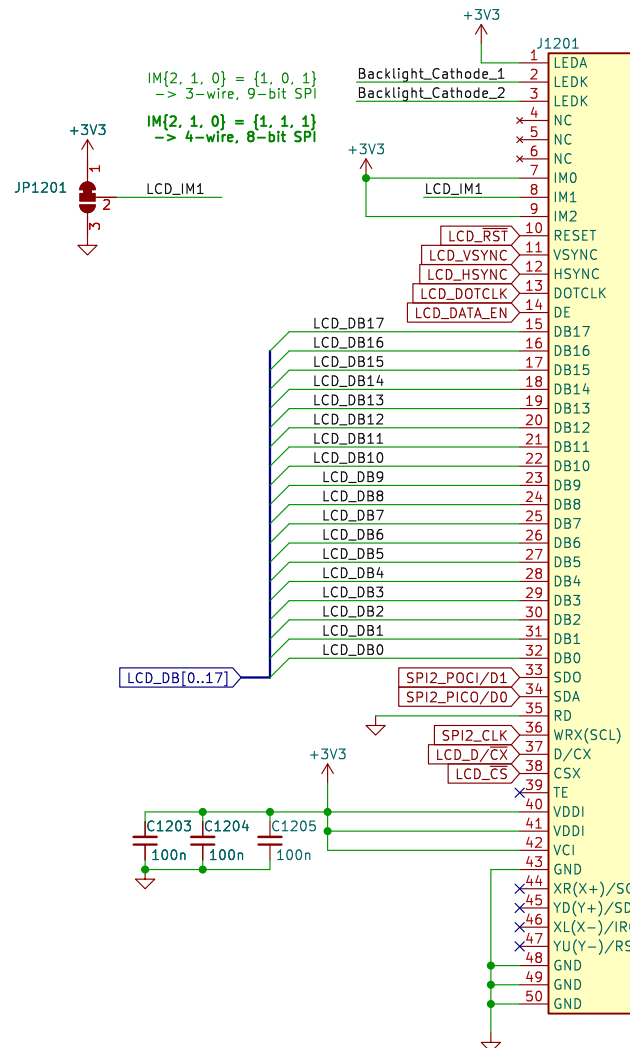


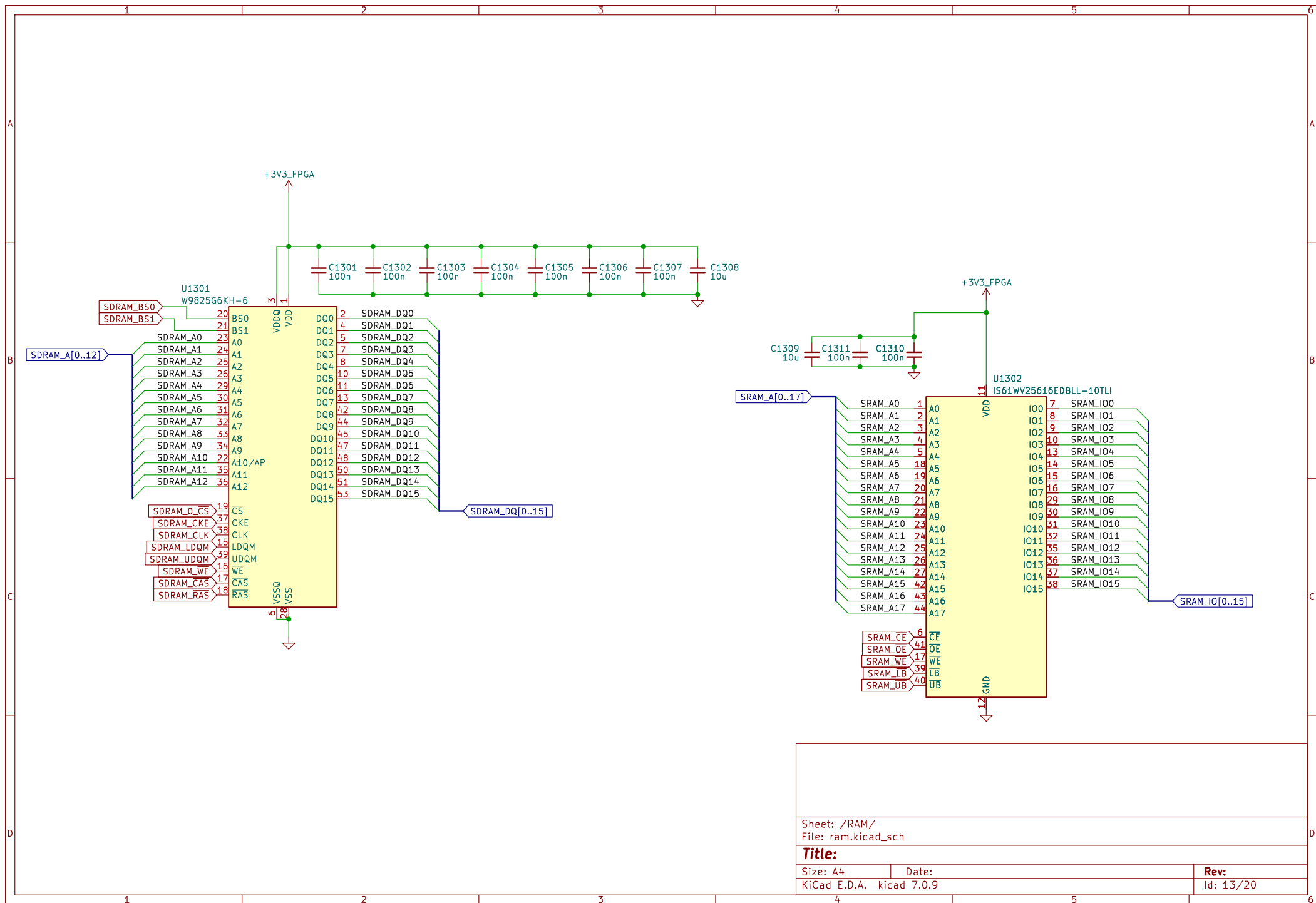




6 LEDs in parallel
 $V_f = 2.9V - 3.2V$
 $I_{max} = 120\text{ mA}$

Each AP2502 channel provides 20mA
 Total of 120mA from the two





Sheet: /RAM/
File: ram.kicad_sch

Title:

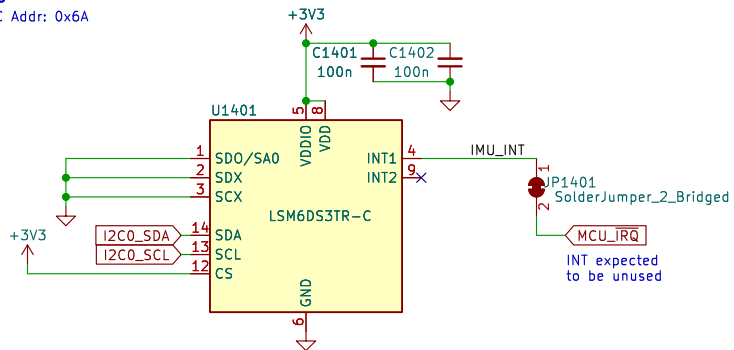
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KiCad E.D.A. kicad 7.0.9

Date:

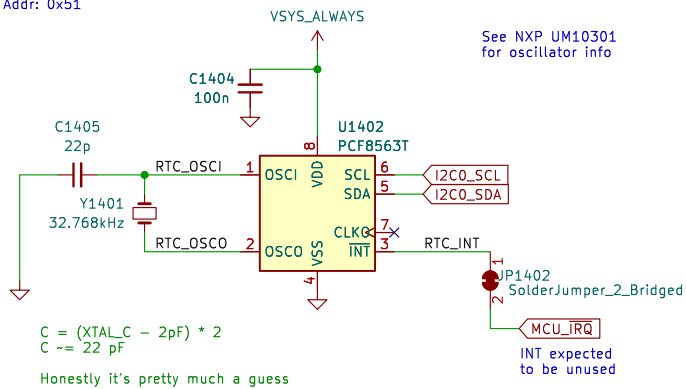
Rev:

Id: 13/20

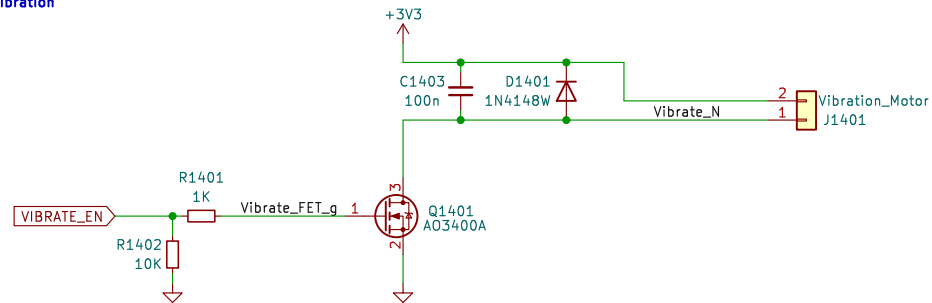
IMU
I2C Addr: 0x6A



RTC
I2C Addr: 0x51



Vibration



Sheet: /Peripherals/
File: peripherals.kicad_sch

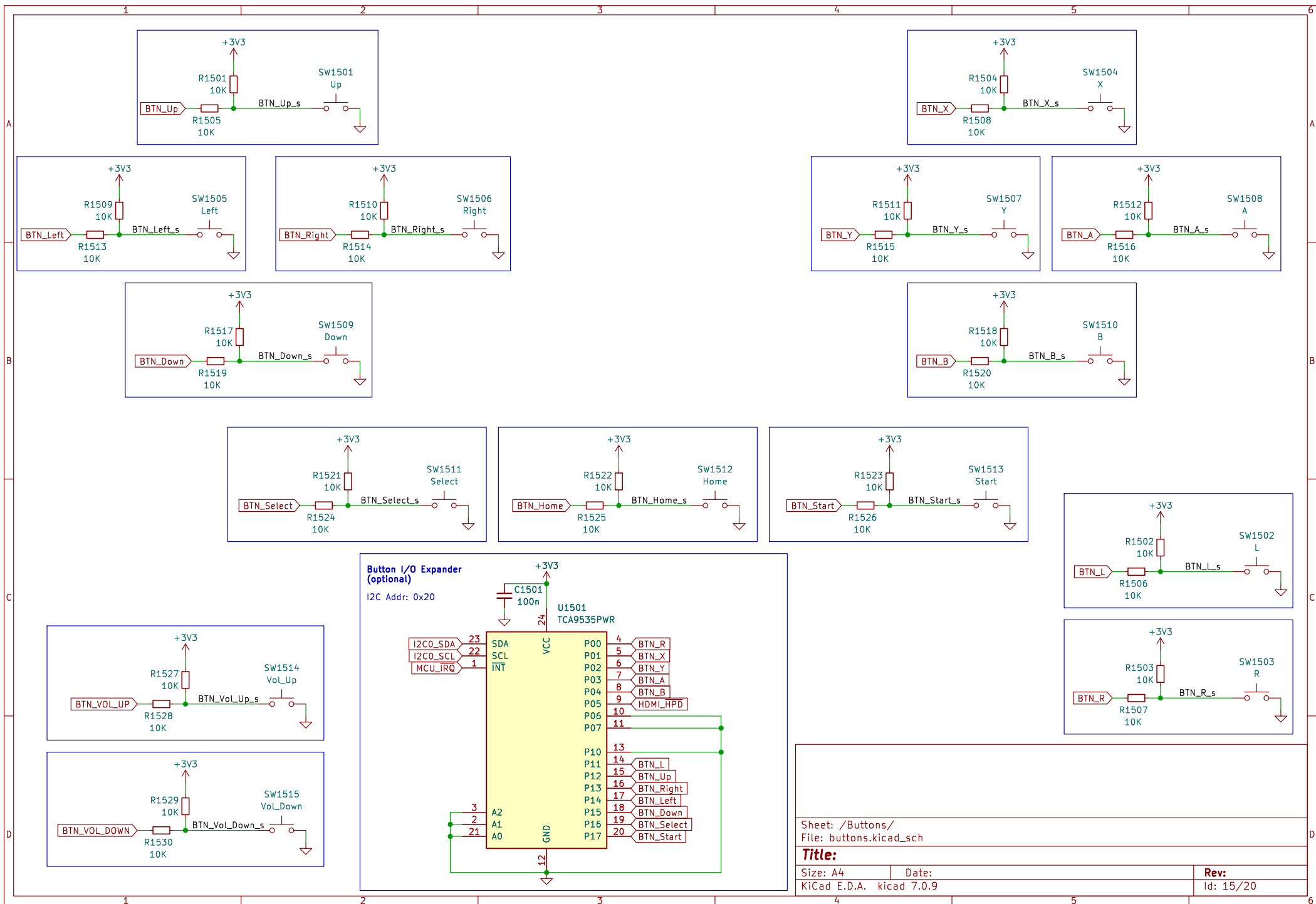
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Size: A4
KiCad E.D.A. kicad 7.0.9

Date:

Rev:

Id: 14/20



Sheet: /Buttons/
File: buttons.kicad_sch

Title:

Size: A4

Date:

KiCad E.D.A. kicad 7.0.9

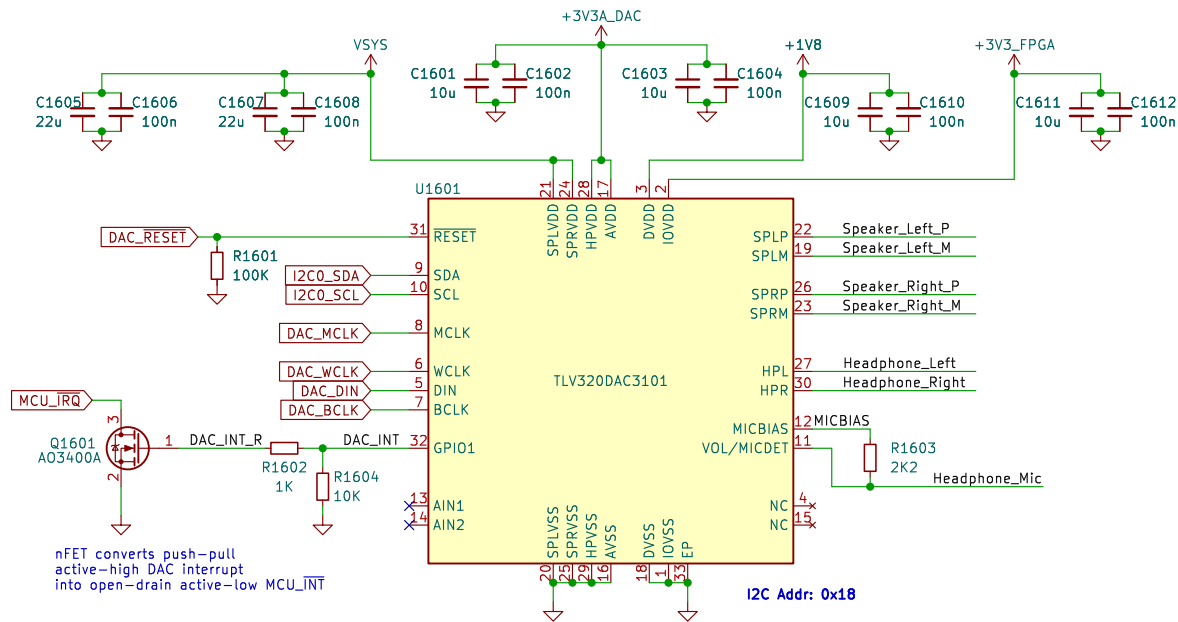
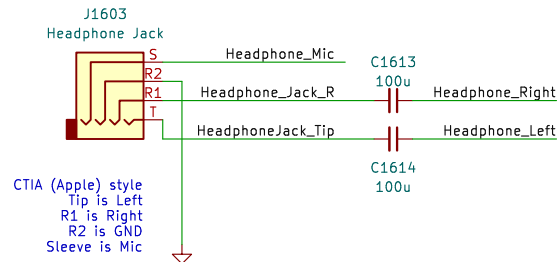
Rev:

Id: 15/20

Test Points

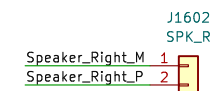
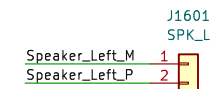


Headphone Jack



Speaker

CMS-23558N-L152B



TODO
check polarity and pins of speaker

Sheet: /Audio/
File: audio.kicad_sch

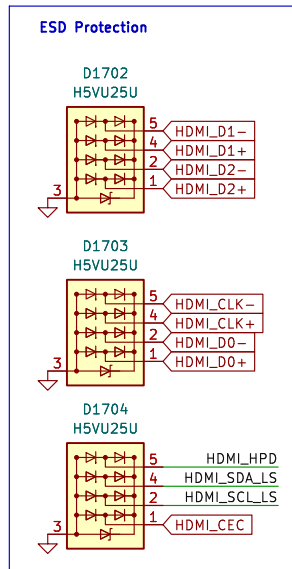
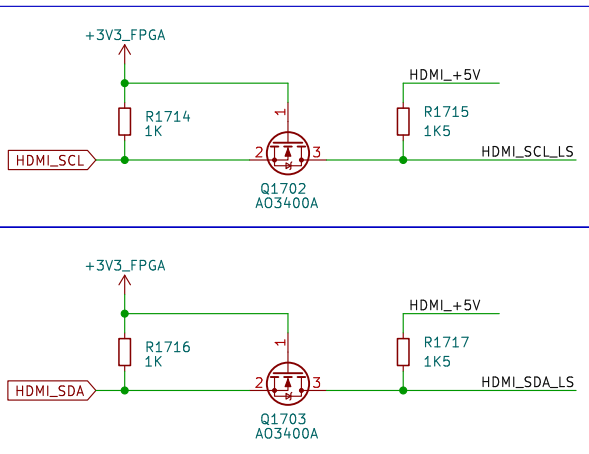
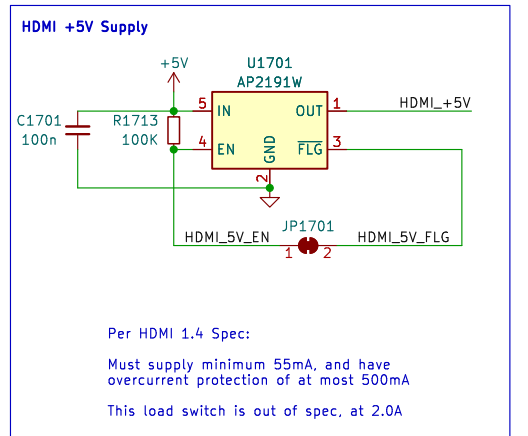
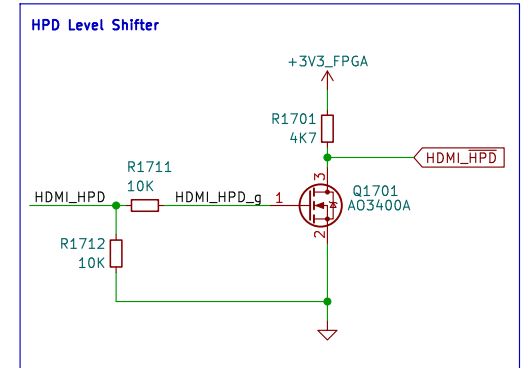
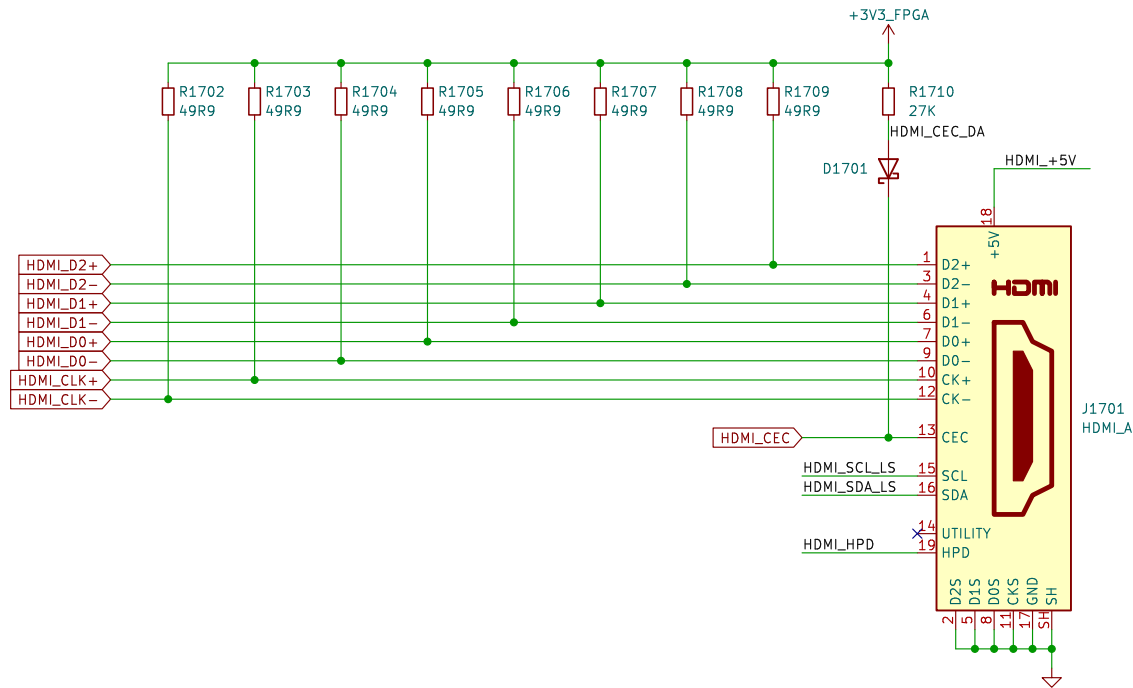
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Size: A4
KiCad E.D.A. kicad 7.0.9

Date:

Rev:

Id: 16/20



Sheet: /HDMI/
File: hdmi.kicad_sch

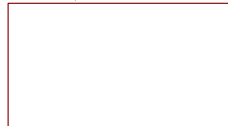
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Date:

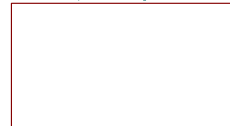
Rev:
Id: 17/20

Game Boy Link



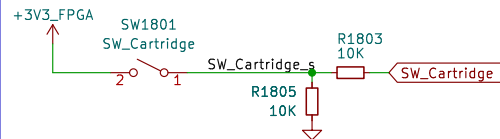
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Game Boy Cartridge

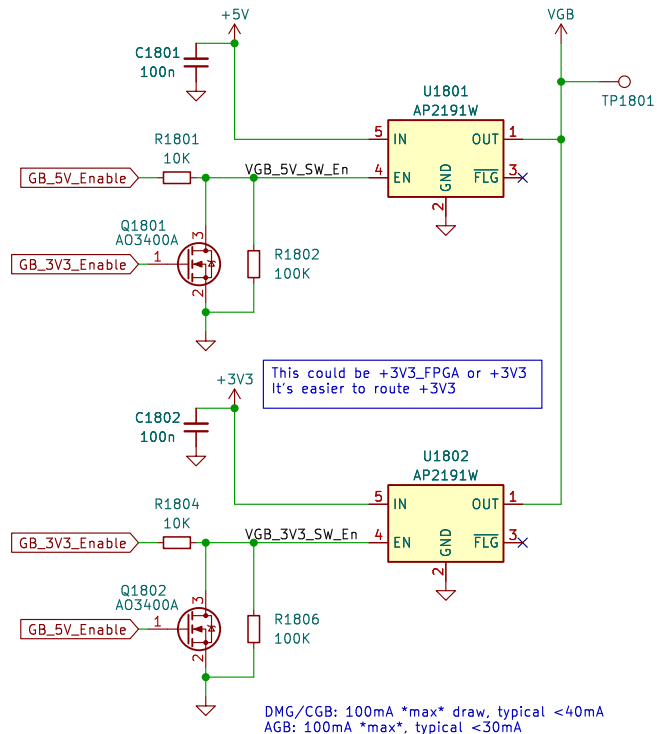


File: gb_cart.kicad_sch

Cartridge Shape Detector



VGB Mux



nFETs ensure enables are mutually exclusive

Sheet: /Game Boy Interfaces/
File: gb_interfaces.kicad_sch

Title:

Size: A4

Date:

KiCad E.D.A. kicad 7.0.9

Rev:

Id: 18/20

